

C949

Data Structures and Algorithms I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An undo feature in a text editor used by 2,000 employees keeps actions on a stack. The program runs push 4, push 7, push 2, pop, push 9, pop. What value is on top now?

- A. 4
- B. 2
- C. 7
- D. 9

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A queue runs enqueue A, enqueue B, enqueue C, dequeue, enqueue D, dequeue. Which item is at the front?

- A. D
- B. A
- C. B
- D. C

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A service searches an unsorted singly linked list of 1,000,000 customer records for a matching name. The records are loaded from a CSV exported nightly, and the server has 64 GB of RAM. What is the worst-case time complexity of the search?

- A. $O(\log n)$
- B. $O(n \log n)$
- C. $O(1)$
- D. $O(n)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An online store's product catalog, updated every Tuesday, holds 1,024 items sorted by SKU, and each record is 2 KB. About how many comparisons does binary search need in the worst case to find an item?

- A. About 10
- B. 32
- C. 1,024
- D. 512

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Binary search on a sorted list needs at most about 20 comparisons. Roughly how many items does the list hold?

- A. 400
- B. 20
- C. About 1,000,000
- D. 40

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A hospital with 7 departments stores employee IDs in a binary search tree by inserting 50, 30, 70, 20, 40, 60, 80. The data refreshes each night. Which traversal prints the IDs in ascending order?

- A. Level-order
- B. Post-order
- C. In-order
- D. Pre-order

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

In the tree built from 50, 30, 70, 20, 40, 60, 80, where does 45 go when inserted?

- A. Left child of 60
- B. Right child of 40
- C. Left child of 40
- D. Right child of 30

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A stadium with 40,000 seats uses a hash table of 10 buckets with $h(k) = k \bmod 10$ and linear probing to store badge numbers. It inserts 23, then 43, then 13 into an empty table. In which bucket does 13 end up?

- A. 4
- B. 5
- C. 3
- D. 13

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A hash table holding 12 entries has a load factor of 0.75. How many buckets does it have?

- A. 12
- B. 16
- C. 4
- D. 9

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A chat app keeps 50,000 messages, averaging 280 characters each, and inserts every new message at the front of the list. Which structure makes each front insert $O(1)$?

- A. A linked list
- B. A hash table, because every operation is always $O(1)$.
- C. An array
- D. A sorted array

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A team lead, writing in Java 21, says: “We define push, pop, peek, and isEmpty with their expected behavior. Later we can build it on an array or a linked list.” What has the team defined?

- A. An abstract data type
- B. An algorithm
- C. A data structure
- D. An object instance

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A nightly job sorts 2 million records that usually arrive already in order. It uses quicksort that always picks the first element as the pivot and runs on a 16-core server. What is the likely running time on this input?

- A. $O(n \log n) / 16$, because work splits across cores.
- B. $O(n)$, because the data is already sorted.
- C. $O(n \log n)$, because quicksort is always $O(n \log n)$.
- D. $O(n^2)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

An undo feature in a text editor used by 2,000 employees keeps actions on a stack. The program runs push 4, push 7, push 2, pop, push 9, pop. What value is on top now?

Target word: top of the stack **Decides:** LIFO: pop removes 2, then 9.

A	Inverted operation	That is the bottom of the stack.
B	Confusion set	Traces it as a queue, where 2 would be at the front.
C	Correct	After removing 2 and then 9, the stack is 4, 7 with 7 on top.
D	Dropped element	Forgets the last pop.

2. Answer D Concept 1 · transfer

A queue runs enqueue A, enqueue B, enqueue C, dequeue, enqueue D, dequeue. Which item is at the front?

Target word: front of the queue **Decides:** FIFO: A leaves, then B leaves.

A	Confusion set	That is stack (LIFO) thinking.
B	Inverted operation	Removes from the back instead of the front.
C	Dropped element	Misses the second dequeue.
D	Correct	Remaining queue is C, D.

3. Answer D Concept 2

A service searches an unsorted singly linked list of 1,000,000 customer records for a matching name. ~~The records are loaded from a CSV exported nightly, and the server has 64 GB of RAM.~~ What is the worst-case time complexity of the search?

Target word: search cost on unsorted linked list **Decides:** Must visit nodes one by one.

A	Dropped element	Binary search needs sorted data with random access.
B	Confusion set	That is the cost of sorting, not searching.
C	Adjacent category	That is array access by index, not search.
D	Correct	Each node may need to be checked.

4. Answer A Concept 3

An online store's product catalog, ~~updated every Tuesday~~, holds 1,024 items sorted by SKU, and ~~each record is 2 KB~~. About how many comparisons does binary search need in the worst case to find an item?

Target word: binary search comparisons **Decides:** $\log_2 1,024 = 10$.

A	Correct	Each comparison halves the range: $2^{10} = 1,024$.
B	Inverted operation	Square root, not logarithm.
C	Wrong stem number	Linear search worst case.
D	Adjacent category	Average cost of linear search.

5. Answer C Concept 3 · transfer

Binary search on a sorted list needs at most about 20 comparisons. Roughly how many items does the list hold?

Target word: solve for n **Decides:** $n \approx 2^{20}$.

A	Inverted operation	Squares 20 instead of raising 2 to the 20th.
B	Wrong stem number	That is the comparison count.
C	Correct	$2^{20} = 1,048,576$.
D	Adjacent category	Doubles linearly instead of exponentially.

6. Answer C Concept 4

A hospital ~~with 7 departments~~ stores employee IDs in a binary search tree by inserting 50, 30, 70, 20, 40, 60, 80. ~~The data refreshes each night.~~ Which traversal prints the IDs in ascending order?

Target word: traversal giving sorted output **Decides:** BST ordering property.

A	Adjacent category	Breadth-first: 50, 30, 70, 20, 40, 60, 80.
B	Inverted operation	Node last: 20, 40, 30, 60, 80, 70, 50.
C	Correct	Left, node, right yields 20, 30, 40, 50, 60, 70, 80.
D	Confusion set	Node first: 50, 30, 20, 40, 70, 60, 80.

7. Answer B Concept 4 · transfer

In the tree built from 50, 30, 70, 20, 40, 60, 80, where does 45 go when inserted?

Target word: insert position **Decides:** $45 < 50$ go left; $45 > 30$ go right; $45 > 40$ go right.

A	Adjacent category	45 is less than the root, so it never enters the right subtree.
B	Correct	Follow $50 \rightarrow 30 \rightarrow 40$, then right into an empty spot.
C	Inverted operation	45 is greater than 40, so it goes right.
D	Dropped element	That spot is taken by 40; keep descending.

8. Answer B Concept 5

A stadium ~~with 40,000 seats~~ uses a hash table of 10 buckets with $h(k) = k \bmod 10$ and linear probing to store badge numbers. It inserts 23, then 43, then 13 into an empty table. In which bucket does 13 end up?

Target word: final bucket of 13 **Decides:** $23 \rightarrow 3$; 43 collides $\rightarrow 4$; 13 collides at 3 and 4 $\rightarrow 5$.

A	Adjacent category	That is where 43 landed.
B	Correct	Buckets 3 and 4 are taken, so 13 probes to 5.
C	Dropped element	Home bucket; ignores collisions.
D	Wrong stem number	The key, not a bucket; the table only has 0-9.

9. Answer B Concept 5 · transfer

A hash table holding 12 entries has a load factor of 0.75. How many buckets does it have?

Target word: solve for buckets **Decides:** Buckets = entries / load factor.

A	Wrong stem number	That is the entry count.
B	Correct	$12 / 0.75 = 16$.
C	Adjacent category	That is the number of empty buckets.
D	Inverted operation	Multiplies 12×0.75 .

10. Answer A Concept 6

A chat app keeps 50,000 messages, ~~averaging 280 characters each~~, and inserts every new message at the front of the list. Which structure makes each front insert $O(1)$?

Target word: structure with $O(1)$ front insert **Decides:** Head insertion only relinks one pointer.

A	Correct	Insert at head just updates the head pointer.
B	Absolute language	Hash tables do not keep order and are not always $O(1)$.
C	Confusion set	All 50,000 elements must shift: $O(n)$.
D	Adjacent category	Still shifts elements on insert.

11. Answer A Concept 7

A team lead, ~~writing in Java 21~~, says: “We define push, pop, peek, and isEmpty with their expected behavior. Later we can build it on an array or a linked list.” What has the team defined?

Target word: name the thing defined **Decides:** Operations and behavior with no fixed implementation.

A	Correct	An ADT specifies behavior, not implementation.
B	Dropped element	An algorithm is a procedure; this is a set of operations on data.
C	Adjacent category	The array or linked list would be the data structure.
D	Confusion set	An instance is one created object at run time.

12. Answer D Concept 8

A nightly job sorts 2 million records that usually arrive already in order. It uses quicksort that always picks the first element as the pivot ~~and runs on a 16-core server~~. What is the likely running time on this input?

Target word: quicksort on sorted input **Decides:** First-element pivot on sorted data makes lopsided partitions.

A	Camouflage	Core count does not change the algorithm’s complexity class.
B	Confusion set	That is insertion sort’s best case.
C	Absolute language	That is the average case, not the worst case.
D	Correct	Each partition removes only one element.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Stack (LIFO) vs. queue (FIFO)	A stack removes the most recently added item; a queue removes the oldest item.	Tracing a stack as if it were a queue.	A sequence of push/pop or enqueue/dequeue steps.
2	Big-O of common operations	Array index access is $O(1)$; searching an unsorted list or linked list is $O(n)$; sorting by comparison is $O(n \log n)$.	Assuming all lookups are $O(1)$ or $O(\log n)$.	"How does the running time grow" for a search on a particular structure.
3	Binary search	Binary search needs sorted, random-access data and takes about $\log_2 n$ comparisons.	Using n or $n/2$ as the comparison count.	A sorted collection size that is a power of two.
4	Binary search tree order	In a BST, left subtree $<$ node $<$ right subtree; in-order traversal visits keys in ascending order.	Confusing in-order with pre-order, or stopping insertion at an occupied spot.	Insert sequence or traversal output for a BST.
5	Hashing, collisions, and load factor	The home bucket is $h(\text{key})$; linear probing steps to the next open bucket; load factor = entries / buckets.	Ignoring collisions or inverting the load factor.	Keys with the same remainder, or entries and bucket counts.
6	Linked list vs. array trade-offs	Inserting at the head of a linked list is $O(1)$; inserting at the front of an array is $O(n)$ because every element shifts.	Thinking arrays are always faster.	Frequent inserts at the front or middle.
7	Abstract data type vs. data structure	An ADT defines operations and behavior; a data structure is one concrete way to implement it.	Calling the interface itself the data structure.	"Could be built with an array or a linked list."
8	Sorting complexity	Merge sort is $O(n \log n)$ in every case; quicksort is $O(n \log n)$ on average but $O(n^2)$ with consistently bad pivots, such as the first element of sorted input.	Saying quicksort is always $O(n \log n)$.	Already-sorted input and a simple pivot rule.

Course scope (public description): Data Structures and Algorithms I covers the fundamentals of dynamic data structures, such as bags, lists, stacks, queues, trees, and hash tables with their associated algorithms. This course discusses object-oriented design and abstract data types as design paradigms. The course emphasizes problem-solving and techniques for designing efficient, maintainable software applications. Students will implement simple applications using the techniques learned.